

Adriana Bariego Quintana

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Personal Statement

I am a physicist specializing in data analysis and programming with a passion for Dark Matter Physics. My work brings astrophysics and particle physics together, using computational tools to explore theoretical models. Beyond research, I engage in science communication, sharing insights through my blog and social media platforms.

Current Professional Situation

Predoctoral Fellowship. Employer: Spanish National Research Council (CSIC) | Department: Astroparticle Physics, Institute of Corpuscular Physics (IFIC). | Start date: 09/03/2023 | Professional category: Predoctoral Contract, Group 2. | Dedication: Full-time.

Main Responsibilities: My work within the KM3NeT collaboration focuses on two primary areas:

- **Data Analysis:** Analyzing experimental data to search for dark matter via Cherenkov light signals produced by neutrinos.
- **Instrumentation and Calibration:** Performing laboratory tasks for calibrating Central Logic Boards (CLBs) to prepare them for integration into the detector, which is still under construction.

Research Projects

In progress:

- **Indirect Dark Matter Searches in KM3NeT** (March 2023 – Present)
 - Research on neutrino-based dark matter detection, focusing on theoretical model studies.
 - Affiliation: IFIC-UV.
 - In progress:
 - **WIMP dark matter searches toward the Galactic Centre with KM3NeT/ORCA** — searching ORCA data for the neutrino signature of dark matter annihilation in the Galactic Centre. Presented at ICRC 2025 (*PoS ICRC2025*, 469).
 - **Boosted dark matter searches with KM3NeT/ORCA** — extending the ORCA dark matter programme to boosted dark matter scenarios. In collaboration with Aaron Vincent's group, Queen's University.
- **Effective Theories in High Energy Physics** (August 2020 – Present)
 - Analyzing gravitational theories and dark matter halo shapes at galactic scales.
 - Affiliation: IPARCOS-UCM, collaboration with Felipe J. Llanes-Estrada.
 - In progress:

- **Formation of the spiral feature in Milky Way angular-momentum space** — N-body simulations of the Galaxy, in collaboration with Jorge Peñarrubia.
- **Dark matter halo shapes from gravitational lensing** — constraining halo geometry with lensing observables, in collaboration with Abigail Belarde.
- Completed:
 - **Dark matter halo shapes across the SPARC galaxies from their rotation curves** — statistical test of 175 rotation curves for a Keplerian taper at large radii. Published in *New Astronomy* **126**, 102537 (2026).
 - **Dark matter halo shape of the Milky Way from stellar-stream torsion** — introduced stream torsion as a local, model-independent probe of halo asphericity. Published in *Astronomy & Astrophysics* **687**, A46 (2024).
 - **Dark matter halo shapes across the SPARC galaxies from their rotation curves** — fitted 175 SPARC rotation curves with multipole halo distributions, finding prolate haloes strongly preferred over spherical. Published in *Physical Review D* **107**, 083524 (2023).

Completed:

- **Direct Dark Matter Detection and Neutrinos** (September 2021 – September 2022)
 - Exploration of beyond-standard-model (BSM) theories for dark matter detection.
 - Affiliation: IFT-UAM.

Research stays and external collaboration.

- **15 Sept - 15 Dec 2025: Queen's University, Kingston, Canada** — three-month research stay with Aaron Vincent's group, on boosted dark matter searches with KM3NeT/ORCA. Funded by a BEFPI/2025 predoctoral mobility grant (Generalitat Valenciana), with additional support from the Faculty of Arts and Science International Research Collaborations Fund (PI: A. Vincent).

Education

PhD in High Energy Physics (Ongoing)

- Institution: IFIC-UV, Spain.
- Specialization: Dark Matter, Astroparticle physics, Data Analysis (March 2023-Ongoing)
- Thesis Title: “Indirect searches for Dark Matter in KM3NeT neutrino experiment with the exploration of new theoretical models”
- Supervisors: S. Rebecca Gozzini, Juan De Dios Zornoza

Master of Science in Theoretical Physics

- **Institution:** IFT-UAM, Spain
- **Specialization:** Particle Physics and Cosmology (2021 – 2022)

- **Thesis Title:** "Dark Matter or Neutrinos: What Are We (Not Yet) Seeing in Direct Detection Experiments?"
- **Supervisors:** David G. Cerdeño, Martín De Los Ríos

Bachelor of Science in Physics

- **Institution:** UCM, Spain
- **Specialization:** Fundamental Physics (2017 – 2021)
- **Thesis Title:** "Galactic Rotation Curves and Anisotropic Distribution of Presumed Dark Matter."
- **Supervisor:** Felipe J. Llanes-Estrada

Continuous and Specialized Training

Physics Courses Received

- **3-4 Aug 2026:** Machine Learning - Intermediate Level | [HSE/IRIS-HEP](#)
- **20-23 Jan 2026:** KM3NeT Bootcamp 2026 | KM3NeT Collaboration
- **5-9 Aug 2024:** Stream2024 Workshop: Stellar Streams and Dark Matter | Durham University
- **25 Jun-5 Jul 2024:** ISAPP PhD School: "Particle Candidates for Dark Matter" | Scuola Galileiana di Studi Superiori, Padova
- **21-22 Mar 2024:** Top 10 Symmetries | Universidad de Valencia
- **13-19 Nov 2023:** ISAPP PhD School: "Exploring the Dark Universe from Particles to Galaxies" | University of Amsterdam, GRAPPA
- **25-27 Oct 2023:** Multidark Workshop for Dark Matter | Universidad Politécnica de Valencia
- **11-14 Oct 2023:** KM3NeT Bootcamp 2023 | KM3NeT Collaboration
- **25-27 Mar 2022:** Lectures on Grand Unified Theories (GUT) | IFIC-UV
- **12-15 Dec 2022:** Madrid Winter Workshop on Theoretical Physics | UCM
- **19-22 Oct 2022:** KM3NeT Bootcamp 2022 | KM3NeT Collaboration
- **12-29 Jul 2022:** IFIC Summer Student Programme: Collaboration with KM3NeT Group | IFIC
- **14-22 Jul 2021:** Phy6cool Summer School for Particle, Astroparticle Physics, and Cosmology | CIEMAT
- **Feb 2020:** Techniques in Scientific Software in Python | IPARCOS-UCM
- **2016:** Astrophysics Course | CUM Laude School

Other Courses Received

- **Oct 2025:** Social Media for PhD Students | Unidad Complutense de Madrid
- **Feb 2023:** Writing and Editing: Editing | University of Michigan – Coursera
- **Feb 2023:** Writing and Editing: Drafting | University of Michigan – Coursera
- **Feb 2023:** Writing and Editing: Structure and Organization | University of Michigan – Coursera
- **Feb 2023:** Writing and Editing: Word Choice and Word Order | University of Michigan – Coursera

- **Jul 2020:** Critical Thinking Course | University of Sydney – Coursera
- **Jul 2020:** First Psychological Aids Course | Johns Hopkins University – Coursera

Skills

Programming and Technical Proficiencies

- **Programming Languages:** Python (extensive use), C++, Mathematica (basic knowledge).
- **Research Tools:** Data analysis, numerical methods, simulation, and theoretical modeling in particle physics and astrophysics.

Soft Skills

- Effective communication and collaboration
- Creativity and problem-solving
- Time and project management

Languages

- **Spanish** (Native)
- **English** (C1)
- **German** (B2)
- **French** (B2)

Scientific production

Publications

1. **"No evidence for Keplerian taper of far-out galactic rotation"** Arxiv: [2507.12120](https://arxiv.org/abs/2507.12120)
 - Authors: A. Bariego-Quintana, F.J. Llanes-Estrada.
 - Published in New Astronomy
<https://www.sciencedirect.com/science/article/pii/S1384107626000205>
2. **"A precis of dark matter for mathematicians"**
 - Authors: Felipe J. Llanes-Estrada, Adriana Bariego-Quintana
 - Published in: mathematicians.Trans. A. Razmadze Math. Inst.178 - 2, pp. 197 - 211. 2024. Link: <https://inspirehep.net/literature/2813503>
3. **"The torsion of stellar streams and the overall shape of galactic gravity's source"**
 - Authors: Adriana Bariego-Quintana, F.J. Llanes-Estrada.

- Published in: A&A, Volume 687, no. A46 (July 2024). DOI: [10.1051/0004-6361/202347502](https://doi.org/10.1051/0004-6361/202347502).
4. **"Measuring the Sterile Neutrino Mass in Spallation Source and Direct Detection Experiments"**
 - Authors: D. Alonso-González, D.W.P. Amaral, A. Bariego-Quintana, et al.
 - Published in: JHEP, 12, no. 096 (2023). DOI: [10.1007/JHEP12\(2023\)096](https://doi.org/10.1007/JHEP12(2023)096).
 5. **"Galaxy Rotation Favors Prolate Dark Matter Haloes"**
 - Authors: A. Bariego-Quintana, O. Manzanilla-Carretero, F.J. Llanes-Estrada.
 - Published in: Phys. Rev. D, 107, no. 8, 083524 (2023). DOI: [10.1103/PhysRevD.107.083524](https://doi.org/10.1103/PhysRevD.107.083524).

Other articles

1. **"How Stellar Stream Torsion may reveal aspherical Dark Matter Haloes "**Arxiv : [2405.09678](https://arxiv.org/abs/2405.09678)
 - Authors: A. Bariego-Quintana, F.J. Llanes-Estrada.

Work presented in national and international conferences

1. **WIMP Dark Matter Searches from the Galactic Centre with KM3NeT/ORCA. (talk)** – Identification of Dark Matter 2026 (Jun 1-5, 2026) | Organizer: University of Zaragoza.
2. **WIMP Dark Matter Searches from the Galactic Centre with KM3NeT. (talk)** – ICRC The astroparticle physics conference, Geneva (Jul 14-24, 2025) | Organizer: University of Geneva.
3. **WIMP Dark Matter Searches in the Galactic Centre with KM3NeT (poster)** – Neutrino 2024, Milan, Italy (Jun 16, 2024) | Organizer: University of Milano – Bicocca | Author: Adriana Bariego-Quintana
4. **How Stellar Stream Torsion May Reveal Aspherical Dark Matter Haloes (poster + proceedings)** – 8th IUPAP International Conference on Women in Physics, Online, India (Jul 2023) | Corresponding Author: Yes | Organizer: Gender in Physics Working Group of Indian Physics Association & Tata Institute of Fundamental Research | Contribution: Scientific Book Chapter
5. **Spiral Galaxy Rotation Favors Prolate Dark Matter Haloes (poster)** – European Astronomical Society Meeting, Online (Jun 27 – Jul 1, 2022) | Organizers: European Astronomical Society | Authors: Felipe Llanes Estrada, Adriana Bariego Quintana, Óliver Manzanilla Carretero
6. **Spiral Galaxy Rotation Favors Prolate Dark Matter Haloes (poster)** – American Physical Society Meeting, Online (Apr 9-12, 2022) | Organizer: American Physical Society | Authors: Felipe Llanes Estrada, Adriana Bariego Quintana, Óliver Manzanilla Carretero
7. **Dark-Matter Halo Shapes from Fits to SPARC Galaxy Rotation Curves (poster + proceedings)** – European Physical Society Conference on High Energy Physics 2021 (EPS-HEP2021), Germany (Sep 2021) | Organizer: Universität Hamburg, DESY

Work presented in seminars, workshops, courses...

1. **Dark Matter Searches with KM3NeT**– Multidark Workshop in Tenerife, Spain (Jun 2026, 17-19) | Organizer: MultiDark - Instituto de Astrofísica de Canarias
2. **Tracing the invisible: From galactic rotations to neutrinos and the shape of dark matter haloes (invited seminar)** – Invited Seminar (Nov 17, 2025) | Organizer: Mc Gill University, Montreal, Canada.
3. **The Unseen Cosmos: Hunting WIMPs in the Abyss** – Astroparticle Seminar (Oct 31, 2025) | Organizer: Mc Donald Institute, Queen's University, Kingston, Canada.
4. **Tracing the invisible: Following Dark Matter from Halo Dynamics to Particle Searches (invited seminar)** – Highly Qualified Personnel Seminar (Oct 29, 2025) | Organizer: Mc Donald Institute, Queen's University, Kingston, Canada.
5. **Unveiling low mass WIMP Dark Matter in the Galactic Centre through KM3NeT (talk)** – Dark Matter 2025, Santander, Spain (June 2-6, 2025) | Organizer: Instituto de Física de Cantabria.
6. **Dark Matter Searches with KM3NeT**– Multidark Workshop in Salamanca, Spain (May, 21-23) | Organizer: MultiDark - Universidad de Salamanca
7. **Dark Matter and Exotics (talk)** – KM3NeT Bootcamp 2024, Erlangen, Germany (Dec 3-6, 2024) | Organizer: KM3NeT-ECAP
8. **WIMP and Exotic Searches with KM3NeT/ORCA Detector (talk)** – RENATA & 21st MultiDark Joint Meeting, Santander, Spain (Oct 9, 2024) | Organizer: Instituto de Física de Cantabria
9. **Unveiling Dark Matter Halo Shapes via Stellar Stream Twisting (talk)** – Stream Workshop 2024, Durham, UK (Aug 5-9, 2024) | Organizer: University of Durham
10. **Sensitivities for DM Search in the Galactic Center with KM3NeT/ORCA-6 (talk)** – ISAPP PhD School in Padova, Italy (Jun 25 – Jul 4, 2024) | Organizer: Scuola Galileiana di Studi Superiori
11. **WIMP Dark Matter Searches in the Galactic Centre with KM3NeT (poster)** – Neutrino Conference 2024, Milan, Italy (Jun 17-21, 2024) | Organizer: University of Milan
12. **Indirect Searches for Dark Matter with the KM3NeT Neutrino Telescope (poster)** – ISAPP PhD School in Texel, Netherlands (Nov 13-19, 2023) | Organizer: University of Amsterdam
13. **Search for Light Dark Matter with KM3NeT/ORCA-115 (talk)** – MultiDark Workshop in Gandía, Spain (Oct 25-27, 2023) | Organizer: MultiDark - Universidad Politécnica de Valencia
14. **The Torsion of Stellar Streams: Shaping Gravity (talk)** – MultiDark Workshop in Gandía, Spain (Oct 25-27, 2023) | Organizer: MultiDark - Universidad Politécnica de Valencia
15. **On the Shape of Dark Matter Haloes (talk)** – FlipPhysics Conference 2022, Online (Mar 21-25, 2022) | Organizer: Instituto de Física Corpuscular
16. **Dark Matter Halo Shapes from Fits to SPARC Galaxy Rotation Curves (talk)** – Kumquat Student Conference 2021, Wrocław, Poland (Jan 16, 2021 – Sep 20, 2021) | Organizer: Wrocław University of Science and Technology

Awards and Recognitions

Grants and funding

- **2025 — Predoctoral mobility grant (BEFPI/2025).** €5,700. Conselleria d'Educació, Cultura, Universitats i Ocupació, Generalitat Valenciana. Funded a three-month research stay at Queen's University, Kingston, Canada.

Awards and honours

- **2023 — Best Poster Award.** 8th International Conference on Women in Physics (ICWIP).
- **2021 — Certificate of Distinction.** Kumquat Student Conference.

Outreach Activities

- Physics Unplugged outreach project (active):

Physics Unplugged is where I explore the messy, fascinating life of doing physics, from research adventures and dark matter analyses to the everyday challenges of being a scientist.

- Dedicated online space: [Outreach in Physics and Sciences](#)
- Instagram: [@physicsunplugged](#)
- YouTube: [@physicsunplugged-abq](#)
- Mentor in steMatEsElla program (2023)

Outreach Organized Events

1. **Participation in guided visit of high-school students to KM3NeT/IFIC lab**
Event: Outreach visit for high school students (February 2026) | Organizer: IFIC/UV | Type: High school student visit.
2. **Seminar on Dark Matter Searches at an outreach event in Canada**
Event: Outreach course for high school students in Mc Donald Institute, Kingston, Canada (Oct 11, 2025) | Organizer: Mc Donald Institute, Queen's University | Type: High school student course.
3. **Seminar on Dark Matter Searches in a physics school**
Event: The Physics Erasmus Summer School, Valencia, Spain (Sep 11, 2024) | Organizer: University of Valencia | Type: Summer School
4. **Tutor for Summer School for Undergraduate Students 2024. Topic: Dark Matter Searches in KM3NeT**
Event: IFIC Summer School, Valencia, Spain (Jul 22, 2024) | Organizer: Instituto de Física Corpuscular (IFIC) | Type: Summer School
5. **Tutor for Summer School for Undergraduate Students 2023. Topic: Dark Matter Searches in KM3NeT**
Event: IFIC Summer School, Valencia, Spain (Jul 10, 2023) | Organizer: Instituto de Física Corpuscular (IFIC) | Type: Summer School
6. **IFIC Expociencia 2023. Open Day Activity: "Observe the Universe from the Depths of the Sea: Neutrino Telescopes"**
Event: Expociencia IFIC 2023, Valencia, Spain (May 6, 2023) | Organizer: University of Valencia | Type: Open Day
7. **Tutor for Summer School for Undergraduate Students. Course on Gauge Symmetries and the Standard Model**

Event: Course in the Interdisciplinary Summer School 2022, Valencia, Spain (Jul 18, 2022) | Organizer: Polytechnic University of Valencia | Type: Summer School

8. **Cycle of Physics Talks from the Sofa for Students (online), Organized by the RSEF-YM Student Delegation**

Event: Physics Talks from the Sofa, Online, Spain (2021) | Organizer: Spanish Royal Society of Physics | Type: Online Conference Series

9. **Talk and Visit to CSIC Facilities, Organized by the RSEF-YM Student Delegation**

Event: Talk and Visit to CSIC, Madrid, Spain (Feb 13, 2020) | Organizer: Spanish Royal Society of Physics | Type: Guided Tour and Outreach Talk

Participation in Scientific Committees

National and International Forums and Committees

1. KM3NeT Group of Early Career Scientists (KM3NeT-ECS) – **Representative**, KM3NeT, 01/09/2024 – present
2. Women in Physics Group (GEMF, Spanish Royal Society of Physics) – **Student board member**, Spanish Royal Society of Physics, Spain, 01/11/2022 – 01/11/2024
3. Complutense Delegation Group of Students GdeE RSEF – EPS Young Minds – **Co-founder and Secretary**, RSEF and Young Minds, Madrid, Spain, 2020 – 2021
4. Complutense Delegation Group of Students GdeE RSEF – **Co-founder and Secretary**, Spanish Royal Society of Physics, Madrid, Spain, 2019 – 2020

Scientific Societies and Professional Associations

1. Women in Physics Group (GEMF, Spanish Royal Society of Physics) – Student Member, Spain, 01/11/2022 – 01/11/2024
2. Student Group (Spanish Royal Society of Physics), Spain, 2019 – 01/11/2024
3. Complutense Delegation Group of Students GdeE RSEF – EPS Young Minds – Affiliated with GdeE RSEF and EPS Young Minds, Spain, 2020 – 2021
4. Complutense Delegation Group of Students GdeE RSEF – Affiliated with the Spanish Royal Society of Physics, Spain, 2019 – 2020
5. Spanish Society of Gravitation and Relativity (SEGRE), Spain, 20/10/2023 – present

Attended Conferences and Workshops

17-19 Jun 2026: Multidark Workshop | Instituto de Astrofísica de Canarias, Tenerife, Spain.

8-12 Jun 2025: ANTARES/KM3NeT Collaboration Meeting | Bratislava, KM3NeT Collaboration

1-5 Jun 2026: Identification of Dark Matter 2026 | University of Zaragoza, Spain

26-30 Jan 2024: ANTARES/KM3NeT Collaboration Meeting | Valencia, KM3NeT Collaboration

20-23 Jan 2026: KM3NeT Bootcamp 2024 | Universidad de Granada, KM3NeT Collaboration

14-24 Jul 2025: ICRC The astroparticle physics conference | University of Geneva, Geneva, Switzerland.

2-6 Jun 2025: Dark Matter 2025 | Instituto de Física de Cantabria, Santander, Spain.

21-23 May 2025: Multidark Workshop | Salamanca University, Salamanca, Spain.

3-6 Dec 2024: KM3NeT Bootcamp 2024 | Erlangen, KM3NeT Collaboration

8-11 Oct 2024: RENATA & 21st MultiDark Joint Meeting | IFCA Cantabria, Santander, Spain

16-22 Jun 2024: Neutrino 2024 | University of Milano, Italy

10-14 Jun 2024: ANTARES/KM3NeT Collaboration Meeting | Texel, KM3NeT Collaboration

12-16 Jan 2024: ANTARES/KM3NeT Collaboration Meeting | Bologna, KM3NeT Collaboration

25-27 Oct 2023: MultiDark Workshop | Gandía, Universidad Politécnica de Valencia

16-20 Oct 2023: ANTARES/KM3NeT Collaboration Meeting | Paris, KM3NeT Collaboration

11-14 Jul 2023: ICWIP Conference 2023

5-9 Jun 2023: ANTARES/KM3NeT Collaboration Meeting | Salerno, KM3NeT Collaboration

9-11 Nov 2022: WOW Physics! 2022 | Goethe University, Frankfurt

23-25 May 2022: MultiDark Workshop | Universidad Autónoma de Madrid, IFT

21-25 Mar 2022: FlipPhysics Conference 2022 | Universidad de Valencia

Aug 2021: International Conference of Physics Students (ICPS2021) | IAPS, Denmark

15-16 Feb 2021: 3rd Edition University Physicathon: "Physics Around The Clock" | University of Zaragoza

Aug 2019: International Conference of Physics Students (ICPS2019) | IAPS, Cologne, Germany.